NAME: KIRTI SINGH
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MCQ Quiz – Deep Learning Fundamentals

1. What is Deep Learning?

A) A type of database
B) A subset of Machine Learning using neural networks with many layers
C) A type of programming language
D) A visualization technique

Answer:B) A subset of Machine Learning using neural networks with many layers

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2. Deep Learning is mainly inspired by which system?

A) Computer hardware
B) Human brain
C) Internet networks
D) Database systems

Answer:B) Human brain

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3. Which of the following is a common application of Computer Vision?

A) Speech recognition
B) Image classification
C) Text translation
D) Email filtering

Answer:B) Image classification

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4. A neural network is made up of small units called:

A) Nodes / Neurons
B) Registers
C) Compilers
D) Arrays

Answer:A) Nodes / Neurons

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5. In a neural network, the input layer is responsible for:

A) Making predictions
B) Receiving input data
C) Updating weights
D) Calculating loss

Answer:B) Receiving input data

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6. Which component connects neurons in a neural network?

A) Bias
B) Weight
C) Activation
D) Loss

Answer:B) Weight

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7. A perceptron is:

A) A simple neural network model used for binary classification
B) A clustering algorithm
C) A database structure
D) A visualization tool

Answer:A) A simple neural network model used for binary classification

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8. A perceptron consists of:

A) Input, weights, activation function, output
B) Input, database, output
C) Input, file system, output
D) CPU, memory, output

Answer:A) Input, weights, activation function, output

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9. The perceptron can solve which type of problems?

A) Non-linear problems
B) Linear classification problems
C) Sorting problems
D) Clustering problems

Answer:B) Linear classification problems

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10. Which layer produces the final prediction in a neural network?

A) Input layer
B) Hidden layer
C) Output layer
D) Activation layer

Answer:C) Output layer

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11. Hidden layers in a neural network are used for:

A) Storing data
B) Learning complex patterns
C) Displaying outputs
D) Reducing data size

Answer:B) Learning complex patterns

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12. Increasing the number of hidden layers generally makes the model:

A) Simpler
B) More powerful at learning complex patterns
C) Slower than traditional ML always
D) Unable to learn

Answer:B) More powerful at learning complex patterns

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13. What is the purpose of an activation function?

A) Store data
B) Introduce non-linearity in the model
C) Remove weights
D) Delete neurons

Answer:B) Introduce non-linearity in the model

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14. Which of the following is a commonly used activation function?

A) ReLU
B) Linear Search
C) Bubble Sort
D) SQL

Answer:A) ReLU

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15. The output range of the Sigmoid activation function is:

A) -1 to 1
B) 0 to 1
C) -∞ to ∞
D) 0 to 10

Answer:B) 0 to 1

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16. What is the purpose of a loss function?

A) Increase training speed
B) Measure error between predicted and actual values
C) Store model weights
D) Normalize data

Answer:B) Measure error between predicted and actual values

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17. Which loss function is commonly used for binary classification?

A) Mean Squared Error
B) Binary Cross Entropy
C) Hinge Loss
D) L1 Loss

Answer:B) Binary Cross Entropy

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18. During training, the goal is to:

A) Maximize loss
B) Minimize loss
C) Ignore loss
D) Double the loss

Answer:B) Minimize loss